

List Types of Authentication. Explain Any One

1. Definition:

Authentication is the process of verifying the identity of a user or system before granting access to resources.

2. Types of Authentication:

- 1. Password-Based Authentication**
 - 2. Token-Based Authentication**
 - 3. Biometric Authentication**
 - 4. Certificate-Based Authentication**
 - 5. Multi-Factor Authentication (MFA)**
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3. Explain Any One (Biometric Authentication):

Biometric Authentication is a method of verifying identity using unique biological characteristics of a person.

Examples:

- **Fingerprint**
- **Face recognition**
- **Iris scan**
- **Voice recognition**

Working:

- 1. User provides biometric input (e.g., fingerprint)**
- 2. System captures and converts it into digital form**
- 3. It is compared with stored biometric data**
- 4. If matched → access granted**
- 5. If not matched → access denied**

Advantages:

- **High security**
- **Difficult to duplicate**
- **No need to remember passwords**

Disadvantages:

- Expensive to implement
 - Privacy concerns
 - Errors may occur in recognition
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4. Diagram:

5. Conclusion:

Authentication is essential for system security. Biometric authentication provides strong protection by using unique human characteristics.